

Exercise I

Let $f : U \rightarrow V$ be a linear map with cokernel $\pi : V \rightarrow V / \text{im } f$, and let X be an arbitrary vector space. Show that:

$$\text{im}(X \otimes f) = \ker(X \otimes \pi).$$

Proof. Since $X \otimes \pi$ is surjective, it suffices to demonstrate that the sequence

$$X \otimes U \xrightarrow{X \otimes f} X \otimes V \xrightarrow{X \otimes \pi} X \otimes (V / \text{im } f)$$

is exact at $X \otimes V$, i.e., that $X \otimes (V / \text{im } f)$ serves as the cokernel of $X \otimes f$.

By the universal property of cokernels, this is equivalent to the statement that for any linear map $\alpha : X \otimes V \rightarrow W$, one has $\alpha \circ (X \otimes f) = 0$ if and only if α factors through $X \otimes \pi$.

This is equivalent to asserting that:

$$\text{Hom}(X \otimes (V / \text{im } f), W) \xrightarrow{(-) \circ (X \otimes \pi)} \text{Hom}(X \otimes V, W)$$

is the kernel of:

$$\text{Hom}(X \otimes V, W) \xrightarrow{(-) \circ (X \otimes f)} \text{Hom}(X \otimes U, W).$$

Utilising the canonical isomorphism $\text{Hom}(A \otimes B, C) \cong \text{Hom}(B, \text{Hom}(A, C))$, it suffices to prove that:

$$\text{Hom}(V / \text{im}(f), \text{Hom}(X, W)) \xrightarrow{(-) \circ \pi} \text{Hom}(V, \text{Hom}(X, W))$$

is the kernel of:

$$\text{Hom}(V, \text{Hom}(X, W)) \xrightarrow{(-) \circ f} \text{Hom}(U, \text{Hom}(X, W)).$$

This follows directly, as $\pi : V \rightarrow V / \text{im}(f)$ is the cokernel of $f : U \rightarrow V$.

In summary,

$$\begin{aligned} & \text{Exactness of } X \otimes U \xrightarrow{X \otimes f} X \otimes V \xrightarrow{\pi} X \otimes (V / \text{im } f) \rightarrow 0 \\ \iff & \text{Exactness of } 0 \rightarrow \text{Hom}(X \otimes (V / \text{im } f), W) \rightarrow \text{Hom}(X \otimes V, W) \rightarrow \text{Hom}(X \otimes U, W) \\ \iff & \text{Exactness of } 0 \rightarrow \text{Hom}(V / \text{im}(f), \text{Hom}(X, W)) \rightarrow \text{Hom}(V, \text{Hom}(X, W)) \rightarrow \text{Hom}(U, \text{Hom}(X, W)) \\ \iff & \text{Exactness of } U \xrightarrow{f} V \xrightarrow{\pi} V / \text{im } f \rightarrow 0. \end{aligned}$$

Exercise II

Let $g : U \rightarrow V$ and $f : X \rightarrow Y$ be surjective linear maps. Show that:

$$\ker(f \otimes g) = \ker(f \otimes U) + \ker(X \otimes g).$$

Proof. Clearly, $\ker(f \otimes U)$ and $\ker(X \otimes g)$ are contained in $\ker(f \otimes g)$, so:

$$\ker(f \otimes g) \supseteq \ker(f \otimes U) + \ker(X \otimes g).$$

Conversely, take any $a \in \ker(f \otimes g)$. Then:

- $(f \otimes U)(a) \in \ker(Y \otimes g) = \text{im}(Y \otimes i_{\ker g})$, where $i_{\ker g} : \ker g \rightarrow U$ is the inclusion;

- There exists $b \in Y \otimes \ker g$ such that $(Y \otimes i_{\ker g})(b) = (f \otimes U)(a)$;
- There exists $c \in X \otimes \ker g$ such that $(f \otimes \ker g)(c) = b$;
- Then $(X \otimes i_{\ker g})(c) \in \ker(X \otimes g)$;
- And $(f \otimes U)(a - (X \otimes i_{\ker g})(c)) = 0$, so $a - (X \otimes i_{\ker g})(c) \in \ker(f \otimes U)$.

Thus:

$$a = (a - (X \otimes i_{\ker g})(c)) + (X \otimes i_{\ker g})(c) \in \ker(f \otimes U) + \ker(X \otimes g).$$

It is advisable to depict this argument with a diagram.

Exercise III

Let $\square$ be a rectangle of dimensions $x_0 \times y_0$, where $x_0, y_0 \in \mathbb{R}$. Suppose it can be subdivided into finitely many rectangles $\{\square_i\}_{i=1}^d$, each of which possesses at least one edge of rational length. Show that $\square$ itself must have at least one rational edge.

Proof. Let the dimensions of $\square_i$ be $a_i \times b_i$. Then, $\sum a_i \otimes_{\mathbb{Q}} b_i = x_0 \otimes y_0$. Let $\pi : \mathbb{R} \rightarrow \mathbb{R}/\mathbb{Q}$ be the canonical projection. By assumption, $x_0 \otimes y_0 \in \ker(\pi \otimes \pi)$. By Exercise II, it follows that $x_0 \otimes y_0 \in \mathbb{Q} \otimes_{\mathbb{Q}} \mathbb{R} + \mathbb{R} \otimes_{\mathbb{Q}} \mathbb{Q}$.

Hence, $x_0 \otimes y_0 = 1 \otimes n + m \otimes 1$ for some $m, n \in \mathbb{R}$. Restrict to the vector space $\text{span}_{\mathbb{Q}}(1, m) \otimes_{\mathbb{Q}} \text{span}_{\mathbb{Q}}(1, n)$, which contains $x_0 \otimes y_0$. If either m or n is rational, then x_0 or y_0 is rational. If both are irrational and $x_0 \otimes y_0 = (\alpha + \beta m) \otimes (\gamma + \delta n)$ has no $m \otimes n$ term, then at least one of x_0 or y_0 must be rational.

Exercise IV

Let $\square$ be a rectangle of size $x_0 \times y_0$ with $x_0, y_0 \in \mathbb{R}$. Suppose that it can be partitioned into finitely many squares. Then $x_0/y_0 \in \mathbb{Q}$.

Proof. Assume without loss of generality that $x_0 = 1$. Then:

$$1 \otimes y_0 = \sum_{i=1}^n d_i \otimes d_i,$$

for some $d_i \in \mathbb{R}$. Let $V = \text{span}(y_0, d_1, \dots, d_n)$. Consider any linear functional $f : V \rightarrow \mathbb{Q}$ such that $f(1) = 1$ and $f(y_0) = -1$. Then:

$$(f \otimes f)(1 \otimes y_0) = -1 < 0 \leq \sum f(d_i)^2,$$

a contradiction.

Exercise V

Show that 祖暅's theorem cannot be proven using only algebraic or combinatorial methods; that is, one cannot divide a tetrahedron into finitely many slices that reassemble into a cube.

Proof. For each edge L_i in a polyhedron, let ℓ_i denote its length, and α_i the associated dihedral angle. Then:

$$\sum \ell_i \otimes_{\mathbb{Q}} \frac{\alpha_i}{\pi}$$

is invariant under finite decomposition and gluing. This sum evaluates to zero for a cube, but is non-zero for a tetrahedron with irrational edges, thereby precluding any such dissection into a cube.